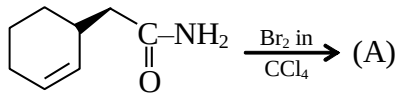
CHEMISTRY

SECTION – I

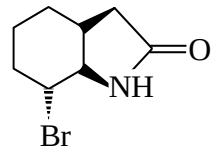
Single Correct Choice Type

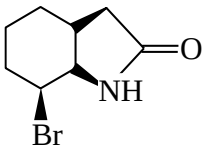
This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

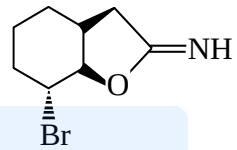
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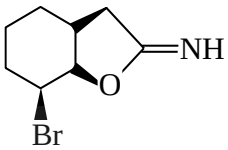
1. 

The product (A) in the given reaction is

(a) 

(b) 

(c) 

(d) 

2. A solution of a monoprotic weak acid has acidity constant K_a. The minimum initial concentration 'c' in terms of K_a, such that the concentration of the undissociated acid can be equated to 'c' within a 10% limit of error, would be

(a) 30 K_a (b) 50 K_a

(c) 70 K_a (d) 90 K_a

3. An optically active substance undergoes first order mutarotation in an aqueous solution. The observed rotation at 0 min, 180 min and ∞ time are 190°, 150° and 30° respectively. The sum of forward and backward rate constants ($k_f + k_r$) would be (log 3 = 0.477, log 2 = 0.3)

(a) 1.57×10^{3} min^{-1} (b) 1.57×10^{-2} min^{-1}

(c) 1.57×10^{-3} min^{-1} (d) 15.7 min^{-1}

4. Which of the following is an incorrect statement? (Atomic numbers: P = 15, S = 16, I = 53)

(a) Axial P–Cl bonds in PCl_5 are longer than equatorial P–Cl bonds.
(b) SF_4 has distorted tetrahedral arrangement.
(c) Bond order of I–I bond in I_3^- is one.
(d) Naphthalene has two types of C–C bond lengths.

5. Which of the following complex represents correct property indicated with it? (Given: atomic number of Fe = 26, Ni = 28)

(a) $[Fe(CO)_5]$: square pyramidal
(b) $[Ni(DMG)_2]$: Colourless
(c) $[Ni(CN)_4]^{2-}$: $\mu = \sqrt{8}$ BM
(d) $[Fe(CN)_5NO]^{2-}$: d^2sp^3 hybridization

6. 0.2063 g of an organic compound (molar mass 168) was heated with sufficient amount of HI and the resulting solution was treated with alcoholic $AgNO_3$ solution. This led to precipitation of 0.8658 g of AgI. The number of methoxy groups in one molecule of the organic compound is/are (Given: atomic mass of Ag = 108, I = 127)

(a) 1
(b) 2
(c) 3
(d) 4

7. Which of the following metal carbide on hydrolysis gives methane?

(a) Mg_2C_3
(b) SiC
(c) Mn_3C
(d) Na_2C_2

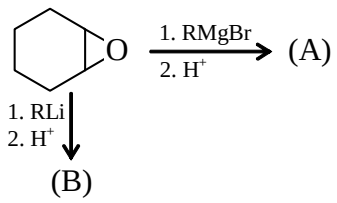
8. Analysis show that nickel oxide consist of nickel ion with 96% ions having d^8 configuration and 4% ions having d^7 configuration. Which among the following best represents the formula of the nickel oxide? (Atomic number of Ni = 28)

(a) $Ni_{1.02}O$
(b) $Ni_{0.96}O$
(c) $Ni_{0.97}O$
(d) $Ni_{0.98}O$

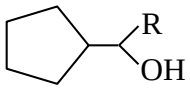
SECTION – II
Multiple Correct Choice Type

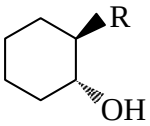
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

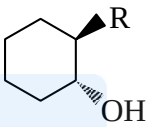
9.

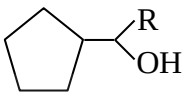


Which of the following is/are correct about structures of compound (A) and (B)?

(a) (A): 

(b) (B): 

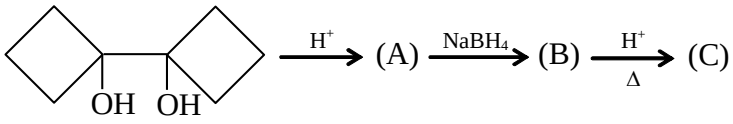
(c) (A): 

(d) (B): 

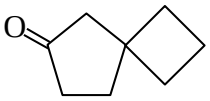
10. Which of the following statement(s) is/are correct for the given reversible process in a state of equilibrium?

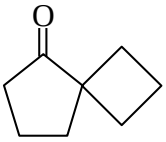
$$2Ag^{+} + 2Cl^{-} \rightleftharpoons Cl_2 + 2Ag$$

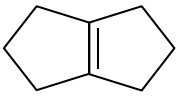
(a) $\Delta G = -2.303\ RT \log K_p$
(b) $\Delta G^\circ = -2.303\ RT \log K_p$
(c) $\Delta G = -2.303\ RT \log K_c$
(d) $\Delta G^\circ = -2.303\ RT \log K_c$

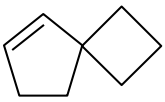
11. 

Which of the following is correct about the products (A), (B) and (C)?

(a) (A): 

(b) (A): 

(c) (C): 

(d) (C): 

12. Two liquids A and B have same molar mass, form a liquid mixture with $X_A = \frac{1}{3}$ and total pressure of 600 mm Hg. The liquid mixture is distilled at 120°C, when $\frac{1}{4}$ of it is distilled. The mole fraction of A in the distillate is 0.75. Which of the following is/are true regarding this mixture? (Given: P_A^o = 300 mm Hg)

(a) Vapour pressure of pure B is 750 mm Hg.

(b) Number of moles of A in the distillate is $\frac{3}{16}$ of the total moles in original solution.

(c) Moles of B in the residue is $\frac{7}{36}$ of the total moles in original solution.

(d) Total vapour pressure of the residue is 662.5 mm Hg.

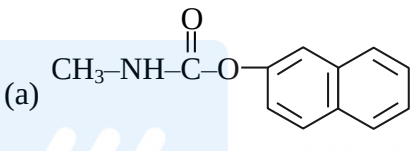
SECTION–III
Comprehension Type

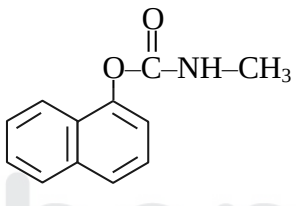
This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

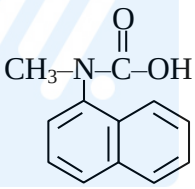
Passage–I

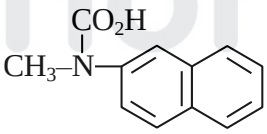
An insecticide SEVIN is made by the reaction of 1-naphthol with methyl isocyanate. A union carbide plant in Bhopal, India, were using this process to make SEVIN for use as an agricultural insecticide. On December 3, 1984, either by accident or by sabotage, a valve was opened that admitted water to a large tank of methyl isocyanate. The pressure and temperatures within the tank rose dramaticity and the pressure relief valves opened to prevent the tank from bursting. A large quantity of methyl isocyanate rushed through the pressure relief valves and the vapours flowed with the breeze into populated areas, killing about 2500 people and injuring many more.

13. The structure of SEVIN is

(a) 

(b) 

(c) 

(d) 

14. The pressure rise in the container (tank) was due to liberation of

(a) NH_3 (b) CH_4

(c) CO_2 (d) CO

15. The temperature rise in the tank was attributed to

(a) Dissolution of NH_3 in water. (b) Dissolution of CO_2 in water.

(c) Dissolution of CH_3–N=C=O in water. (d) Dissolution of CH_3NH_2 in water.

Passage–II

Successive effusion is used for isotopic separation from a gaseous mixture. Let there are two (isotopic) gases A and B present in a container having n_{A_o} and n_{B_o} moles respectively. Effusing this mixture into another container enriches the mixture in that gas which effuses faster. Let, after the first effusion stage the number of moles of A and B are n_{A_1} and n_{B_1} respectively. If there are 'm' number of effusion stages used and finally the number of moles of A and B in the last container is n_{A_m} and n_{B_m} respectively.

The ratio of relative concentration of given (isotopic) species after processing to its relative concentration before processing is referred as separation factor (f).

$$f = \frac{n_{A_m} / n_{B_m}}{n_{A_o} / n_{B_o}}$$

For a single stage effusion process, the maximum (ideal) separation factor (f') is the ratio of rate of effusion of isotopic molecules to the ratio of its relative concentration before processing. This can be determined using Graham's law.

$$f' = \frac{n_{A_1} / n_{B_1}}{n_{A_o} / n_{B_o}} = \frac{r_A / r_B}{n_{A_o} / n_{B_o}}$$

16. Natural uranium consists of 0.1% of U^{235} by mole. The uranium ore is reacted with fluorine to form volatile uranium hexafluoride. The number of effusion steps required to achieve a concentration of 99.9% of $^{235}UF_6$, would be (atomic mass of F = 19)

[Given: log 352 = 2.546, log 349 = 2.542]

(a) 300 (b) 3,000

(c) 30,000 (d) 3,00,000

17. The number of effusion stages needed to enrich initial molar ratio of H_2 and He from 2 : 1 to 64 : 1 by means of successive effusion, would be

(a) 4 (b) 8

(c) 10 (d) 14

18. A sample of hydrogen contains 90% of D_2 and rest is H_2. This mixture is subjected to isotopic enrichment by successive effusion technique. The percentage of H_2 in the mixture after 18 effusion steps is approximately

(a) 100 % (b) 98 %

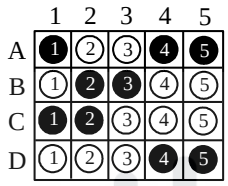
(c) 96 % (d) 93 %

SECTION–IV

Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

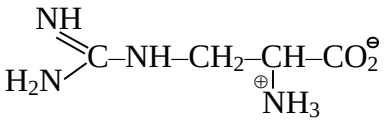
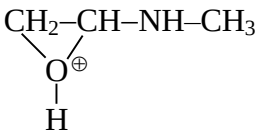
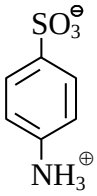
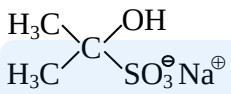
If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

Column I	Column II
(A) $[Co(NO_2)_2(H_2O)_2(NH_3)_2]NO_3$	**1.** Number of stereoisomers = 6
(B) $[Ni(en)_3]Br_3$	**2.** Linkage isomerism
(C) $[Co(NH_3)_3(py)_3]Br_2$	**3.** Ionization isomerism
(D) $[Pt(en)(SCN)_2]\ (NO_3)_2$	**4.** Optical isomerism
	5. Geometrical isomerism

2.

Column I	Column II
(A) 	**1.** Show acidic behaviour towards water.
(B) 	**2.** Show basic behaviour towards water.
(C) 	**3.** Show resonance
(D) 	**4.** Zwitter ion **5.** Crystalline salt

mathongo